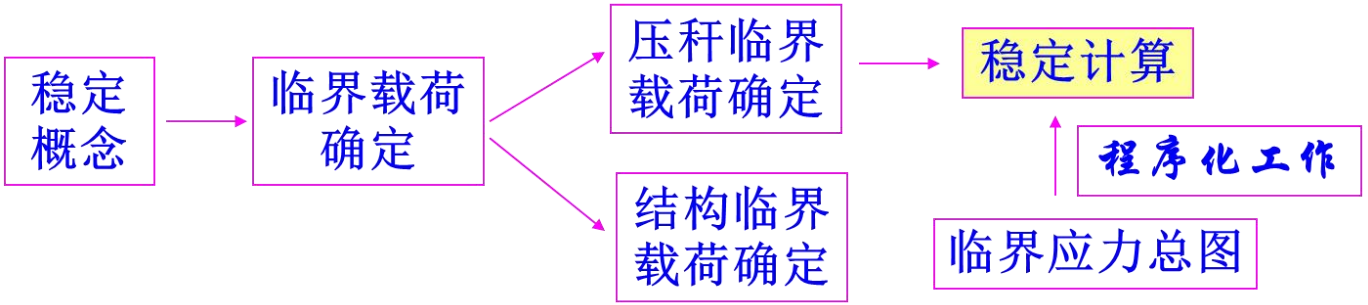
第二十二讲
压杆稳定延伸与拓展

作业：11-14,17

第十一章　压杆稳定

1、静不定结构中的压杆
2、极限分析
3、实际压杆与偏心压缩失稳
4、结构的临界载荷
5、卷尺和塔吊臂承载能力分析

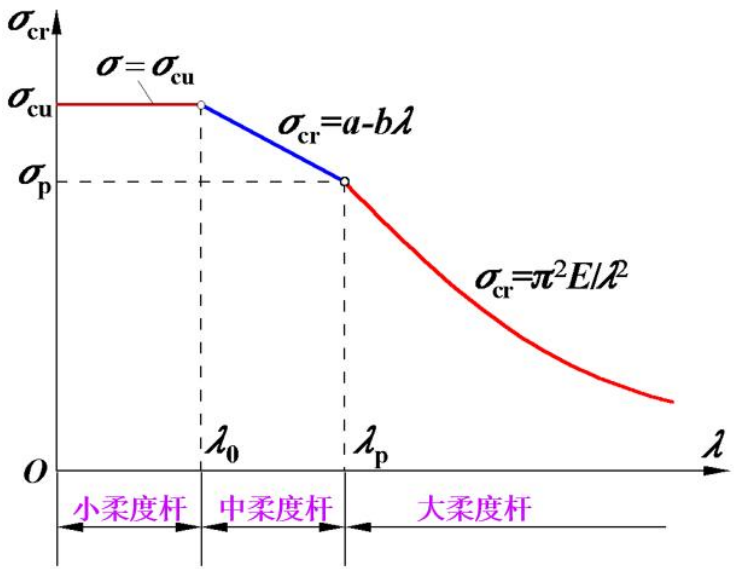


稳定问题回顾

稳定概念

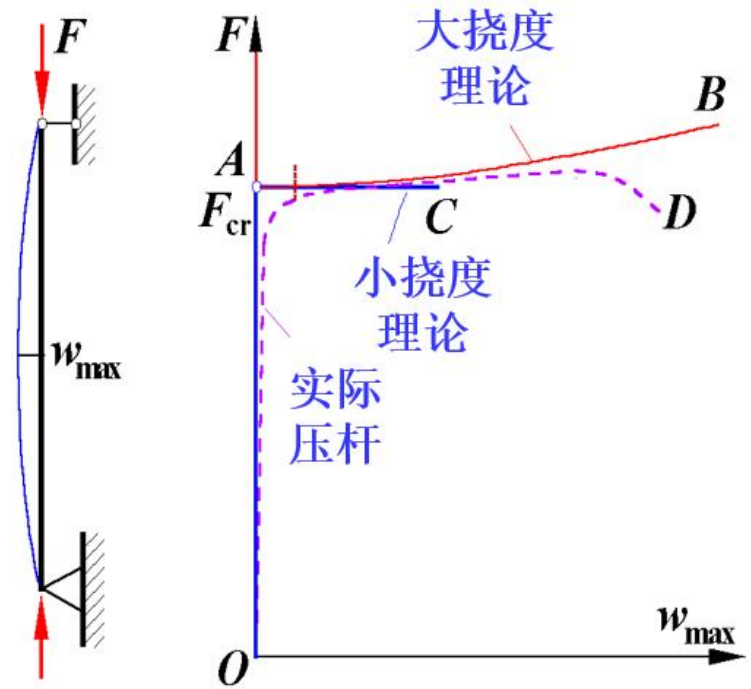
直线平衡状态 → 扰动 → 微小偏离δ 邻近状态 → 平衡 → 临界载荷

临界应力总图



柔度概括了杆长、截面、支持的影响，体现了杆件抗失稳能力

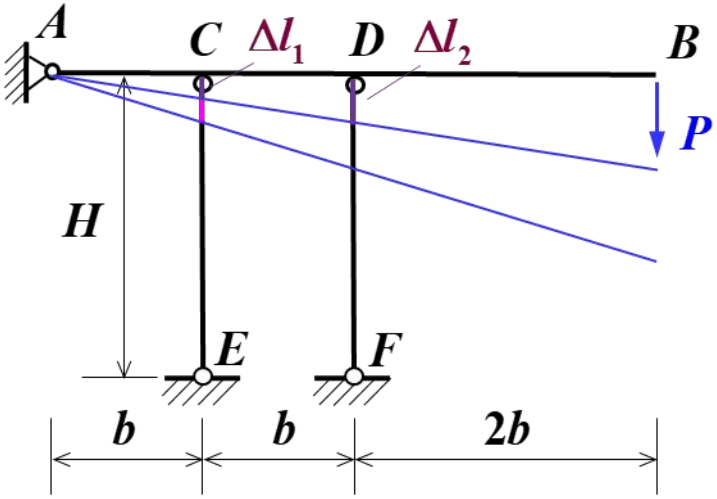
基于小挠度理论的随遇平衡



1、静不定结构中的压杆

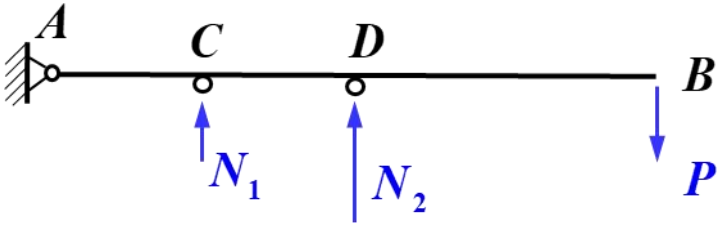
例11-11 图示结构，*AB*为刚性杆，*CE*和*DF*均为大柔度杆（标记为1杆和2杆），*EI*相同且为常数。求许用载荷[*P*]。

解：



$$\left.\begin{aligned} &bN_1 + 2bN_2 = 4bP \\ &2\Delta l_1 = \Delta l_2 \\ &\Delta l_1 = \frac{N_1 H}{EA} \\ &\Delta l_2 = \frac{N_2 H}{EA} \end{aligned}\right\} \quad P = \frac{5}{8}N_2$$

1) 当2杆失稳， $N_2 = N_{cr}$， $P = \dfrac{5\pi^2 EI}{8H^2}$



2) 继续加载， $N_2 = N_{cr}$， $N_1 \uparrow$

当$N_1 = N_{cr}$ $\quad [P] = \dfrac{3}{4}N_{cr} = \dfrac{3\pi^2 EI}{4H^2}$ $\quad$ *Timoshenko* 解

1、静不定结构中的压杆

例11-11 图示结构，*AB*为刚性杆，*CE*和*DF*均为大柔度杆（标记为1杆和2杆），*EI*相同且为常数。求许用载荷[*P*]。

解： *Timoshenko* 解

$$[P]=\frac{3\pi^2 EI}{4H^2}$$

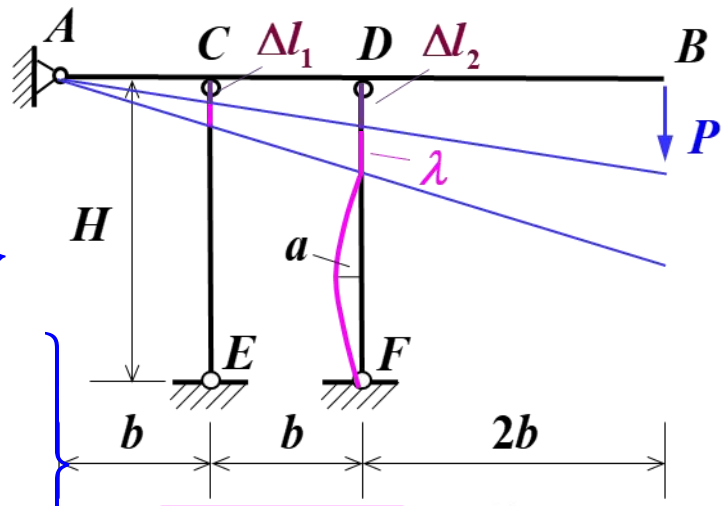
3) 继续加载， 2杆压弯组合

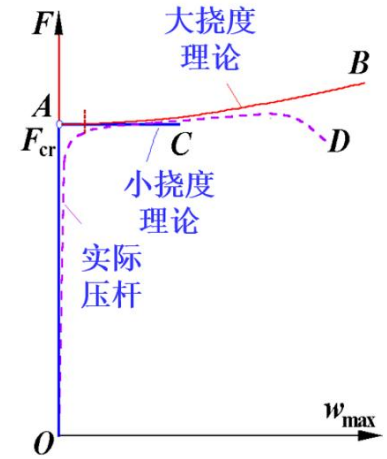
$$w(x)=a\sin\frac{\pi x}{H}$$

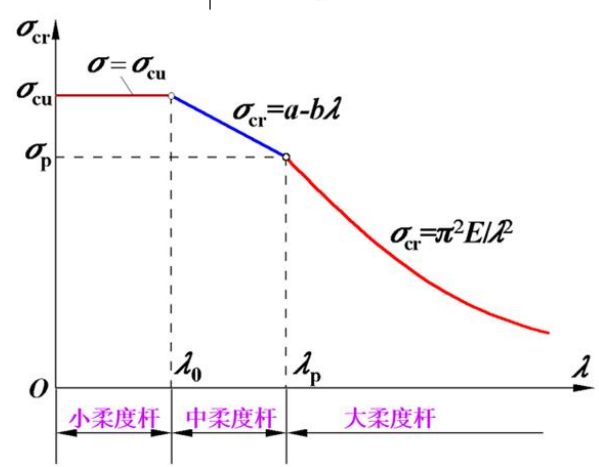
$$\lambda=\frac{N_{cr}H}{EA}=\frac{1}{2}\int_0^H (w')^2\,\mathrm{d}x$$

见§13-8，式（13-31）

$$a=2\sqrt{\frac{I}{A}}\qquad \sigma_{\max}=\sigma_{cr}+\frac{N_{cr}a}{W}$$



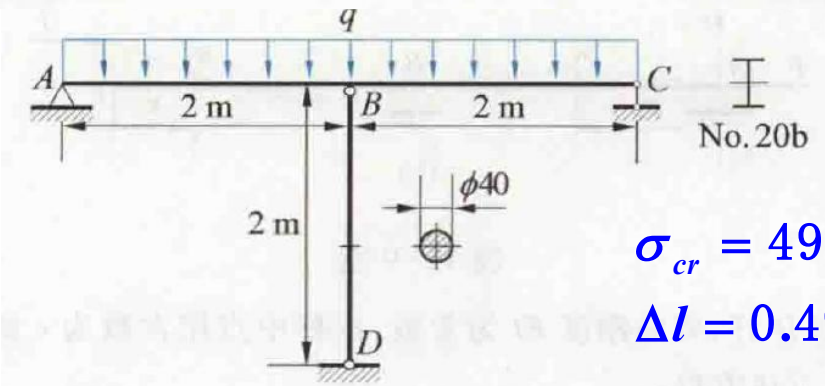




孟庆春，考虑稳定性的结构的失效判据，力学与实践，1995(5):67

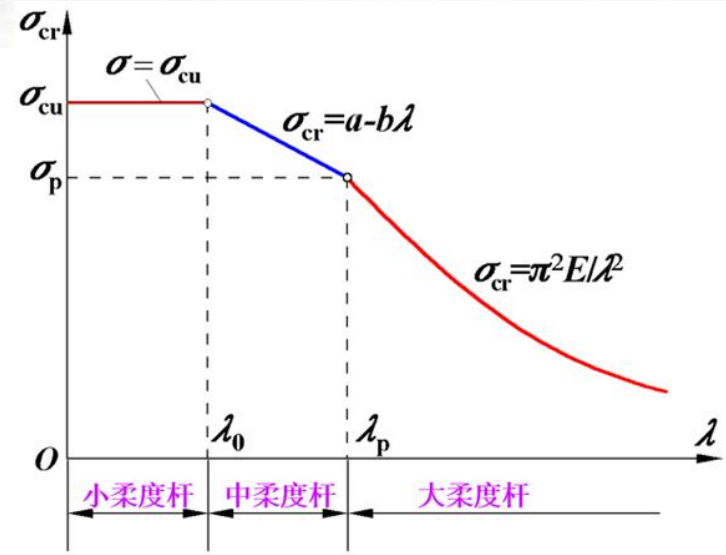
1、静不定结构中的压杆

11-13　图示结构，由横梁 AC 与立柱 BD 组成，材料均为低碳钢，弹性模量 $E=200$ GPa，比例极限 $\sigma_p=200$ MPa。试问当载荷集度分别为 $q=20$ N/mm 与 $q=26$ N/mm 时，截面 B 的挠度分别为何值。



$\sigma_{cr}=49.4$ MPa

$\Delta l=0.4936$ mm



1) $q=20$ N/mm，　σ_{BD}=38.6 MPa

2) $q=40$ N/mm，　BD杆弯曲

$$\left.\begin{aligned} w(x)&=a\sin\frac{\pi x}{l_{BD}} \\ w_B-\Delta l&=\frac{1}{2}\int_0^{l_{BD}}(w')^2\,\mathrm{d}x \end{aligned}\right\}\quad a=88.3\text{ mm}$$

$\sigma_{BD}=49.4+871.2=920.6$ MPa

3) $q=26$ N/mm，　a=15.7 mm

$\sigma_{BD}=49.4+154.8=204.2$ MPa

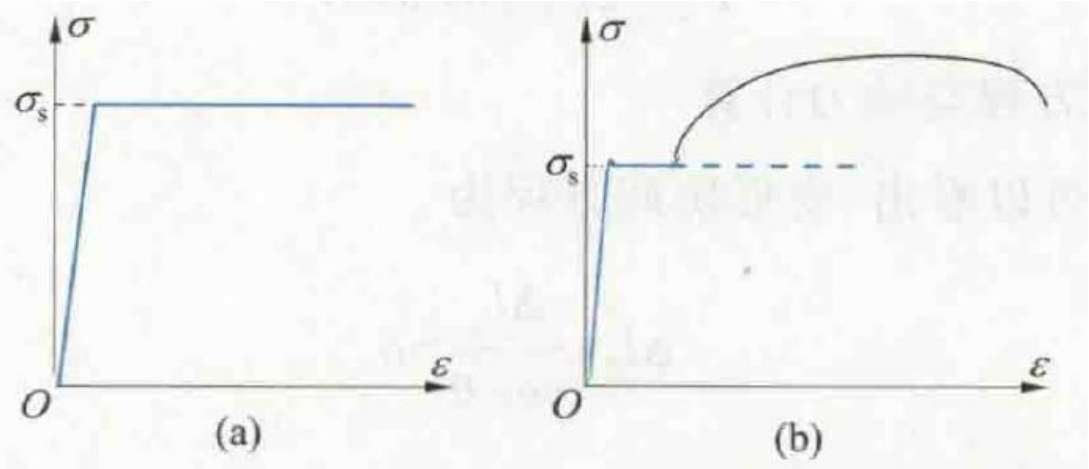
梁的强度裕度大，q=40N/mm时，压杆在压弯组合下破坏。

2、极限分析

§3-7，第十九章

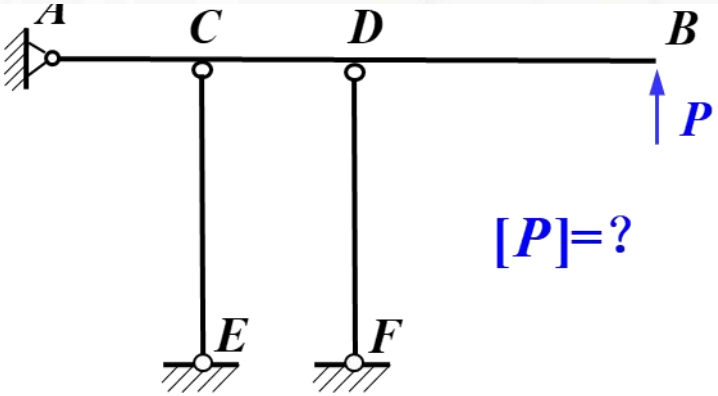
理想弹塑性本构模型

$$\begin{cases} \sigma = E\varepsilon, & \varepsilon < \dfrac{\sigma_s}{E} \\ \sigma = \sigma_s, & \varepsilon \geq \dfrac{\sigma_s}{E} \end{cases}$$



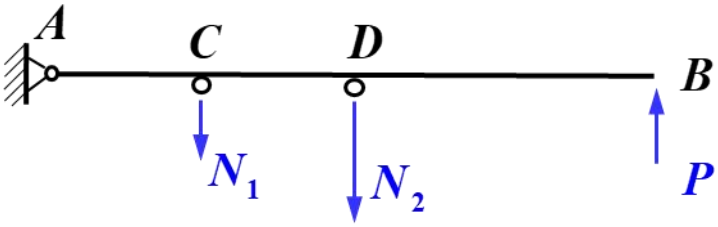
极限载荷

允许构件产生塑性变形，结构所能承受的最大载荷。



$$bN_1 + 2bN_2 = 4bP$$

$$[P] = \frac{3}{4}A\sigma_s$$



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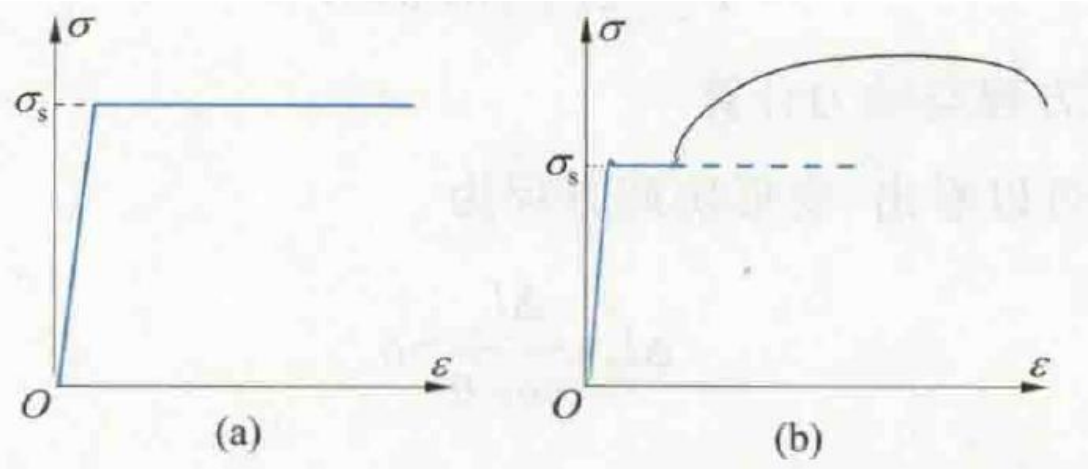
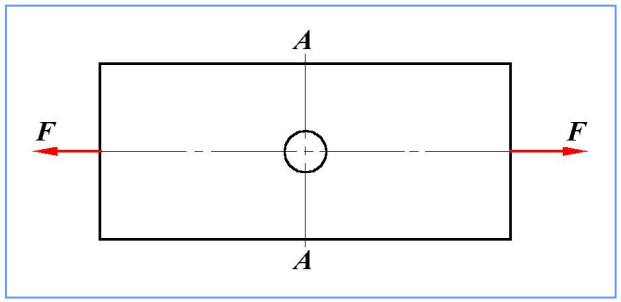
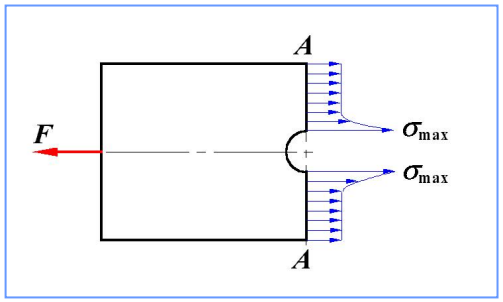
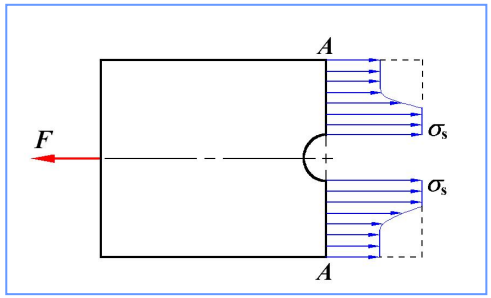
$$\begin{cases} \sigma = E\varepsilon, & \varepsilon < \dfrac{\sigma_s}{E} \\ \sigma = \sigma_s, & \varepsilon \geq \dfrac{\sigma_s}{E} \end{cases}$$



孔边应力集中







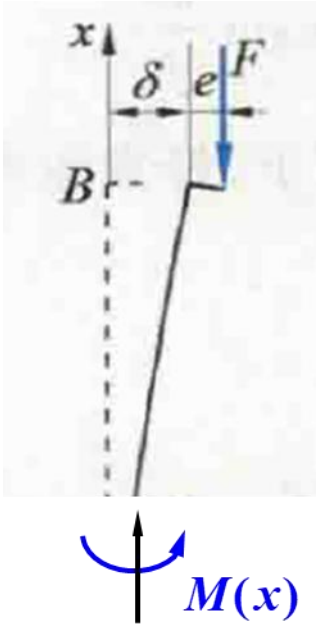
孔边应力集中不影响静强度。

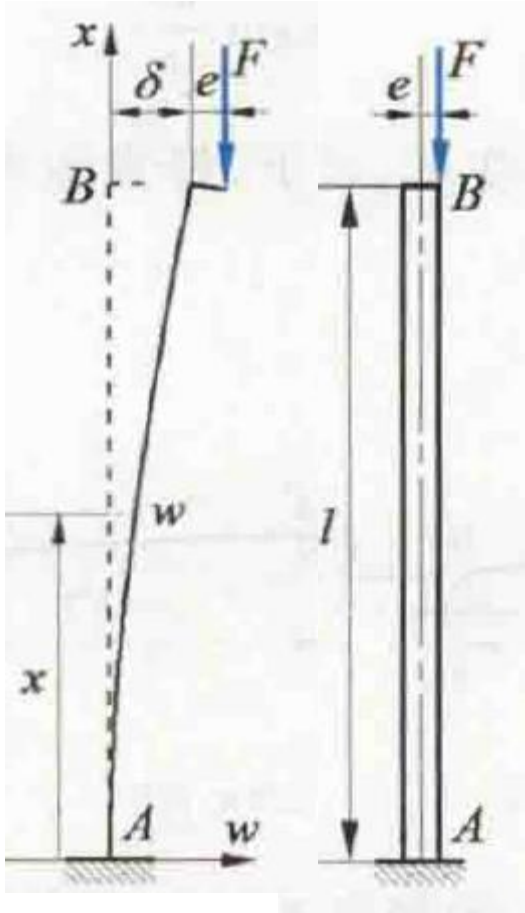
3、实际压杆与偏心压缩失稳

例11-12（习题11-24）

*11-24　图 a 所示细长直杆 AB，顶端 B 承受偏心距为 e 的载荷 F 作用。设横截面的面积为 A，弯曲刚度均为 EI，试求截面 B 的挠度与杆内横截面上的最大正应力。

解： 邻近状态平衡





$$M(x) = F(\delta + e - w)$$

$$\frac{d^2 w}{d x^2} + k^2 w = k^2(\delta + e), \quad k^2 = \frac{F}{EI}$$

$$w(x) = A\sin kx + B\cos kx + \delta + e$$

$$\begin{cases} x = 0, \ w = 0 \\ x = 0, \ w' = 0 \\ x = l, \ w = \delta \end{cases} \Rightarrow \begin{cases} B + \delta + e = 0 \\ A = 0 \\ B\cos kl + \delta + e = \delta \end{cases}$$

$$w(x) = B(\cos kx - 1)$$

3、实际压杆与偏心压缩失稳

*11-24 图 a 所示细长直杆 AB，顶端 B 承受偏心距为 e 的载荷 F 作用。设横截面的面积为 A，弯曲刚度均为 EI，试求截面 B 的挠度与杆内横截面上的最大正应力。

解：

$$\begin{cases} B+\delta+e=0 \\ B\cos kl+e=0 \end{cases}$$

$$w(x)=B(\cos kx-1)$$

1) $e=0 \quad \Rightarrow \quad \cos kl=0$

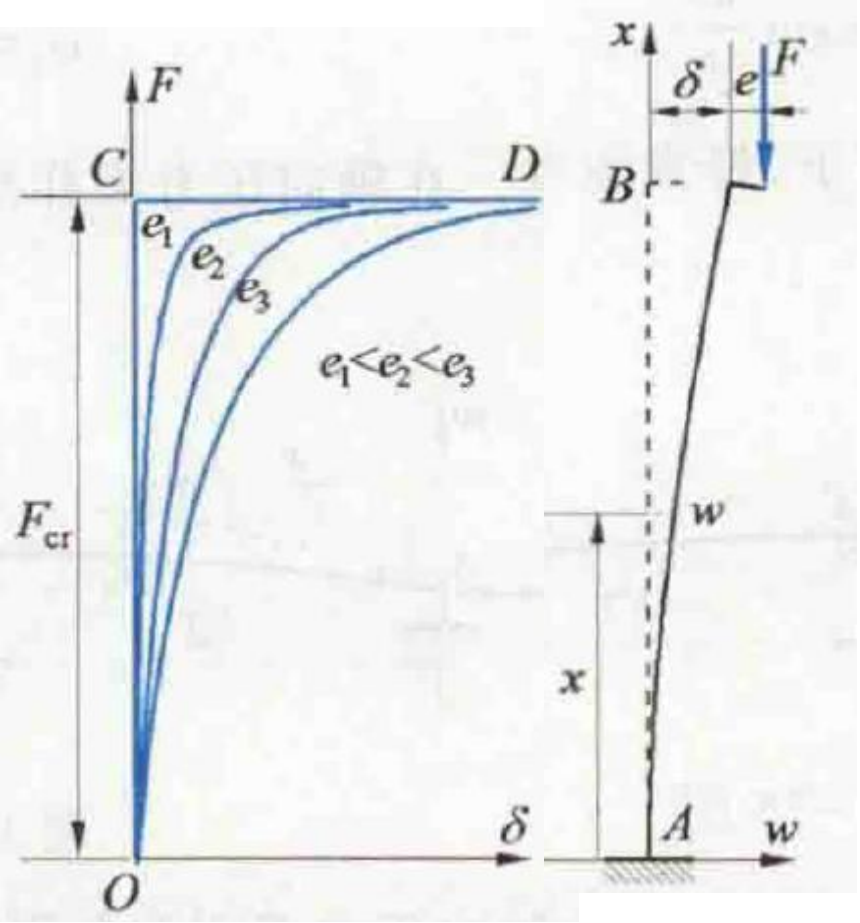
$$kl=\frac{\pi}{2},\frac{3\pi}{2},\cdots \qquad F_{cr}=\frac{\pi^2 EI}{(2l)^2}$$

2) $e\neq 0 \quad \Rightarrow \quad w(x)=\dfrac{e}{\cos kl}(1-\cos kx)$

$$if \quad F\to F_{cr}, \quad kl\to\frac{\pi}{2}, \quad \cos kl\to 0,$$

$$then \quad w_{\max}=w(l)\to\infty$$

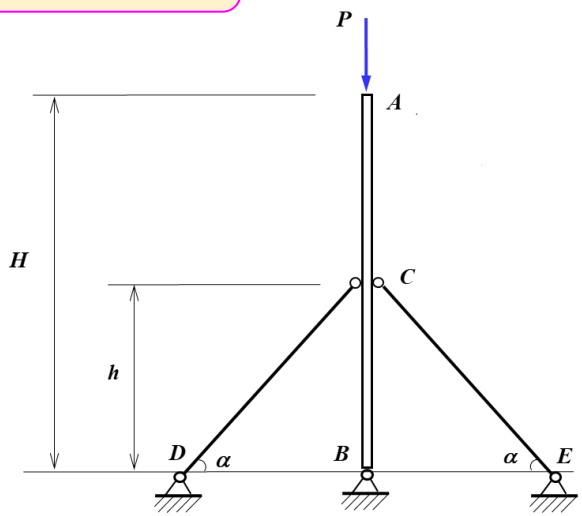
压杆失稳



3、结构的临界载荷

邻近状态的平衡

例11-13 图示竖直刚杆AB，受压力P，用柔索CD和CE拉紧。二柔索相同，长l，刚度EA，预伸长δ_0。求P_{cr}.



解： 设A端偏离δ

$$\Delta l_1 = \Delta l_2 = \frac{h\cos\alpha}{H}\delta$$

$$N_1 = \frac{EA}{l}(\delta_0 + \Delta l_1)$$

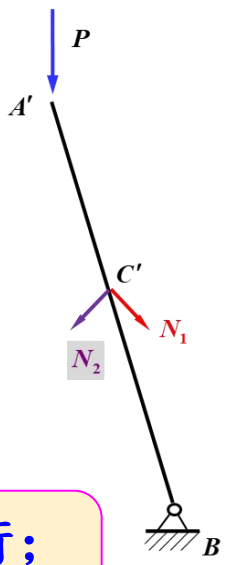
$$N_2 = \frac{EA}{l}(\delta_0 - \Delta l_2)$$

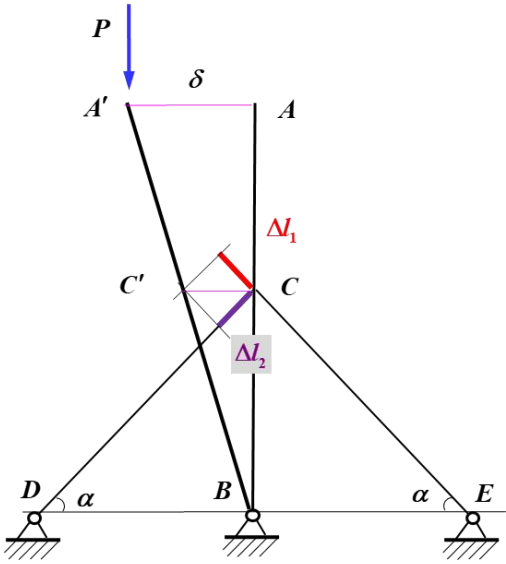
1) $\delta_0 < \Delta l_2$， $P\delta = N_1 h\cos\alpha$

$$P_{cr1} = \frac{EAh}{H}\sin\alpha\cos^2\alpha\left(1+\frac{\delta_0}{\Delta l_2}\right)$$

2) $\delta_0 > \Delta l_2$， $P\delta + N_2 h\cos\alpha = N_1 h\cos\alpha$

$$P_{cr2} = \frac{2EAh}{H}\sin\alpha\cos^2\alpha$$





$P_{cr1} < P_{cr2}$

柔索预紧力提高临界载荷；
扰动过大，临界载荷降低

3、结构的临界载荷

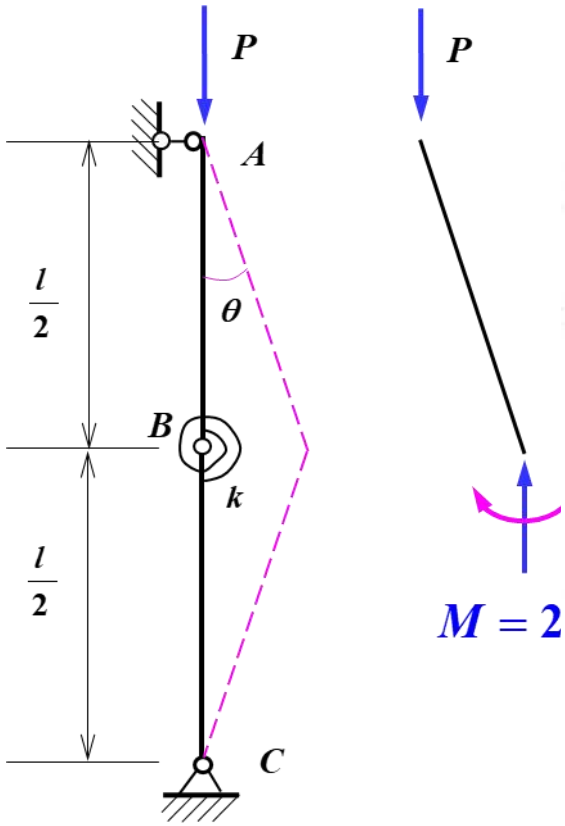
邻近状态的平衡

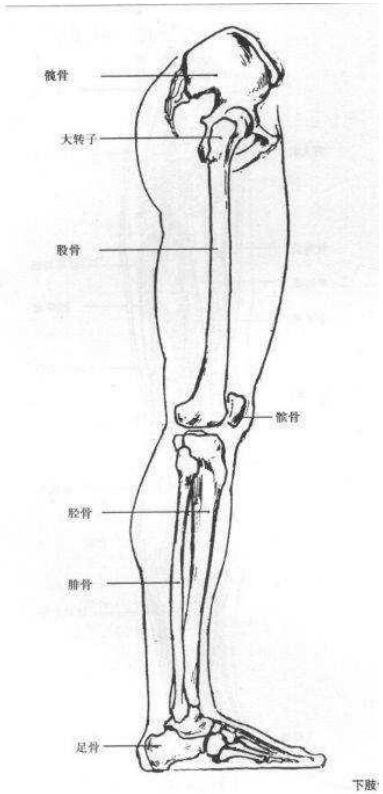
例11-14 图示刚杆AB和BC用扭转弹簧连接。杆长l，轴力P。求弹簧刚度系数k。

解： 设刚杆偏转θ

$$P\frac{l}{2}\theta = 2\theta k$$

$$k = \frac{Pl}{4}$$





大腿绷紧，提高膝关节扭转刚度，防失稳

3、结构的临界载荷

邻近状态的平衡

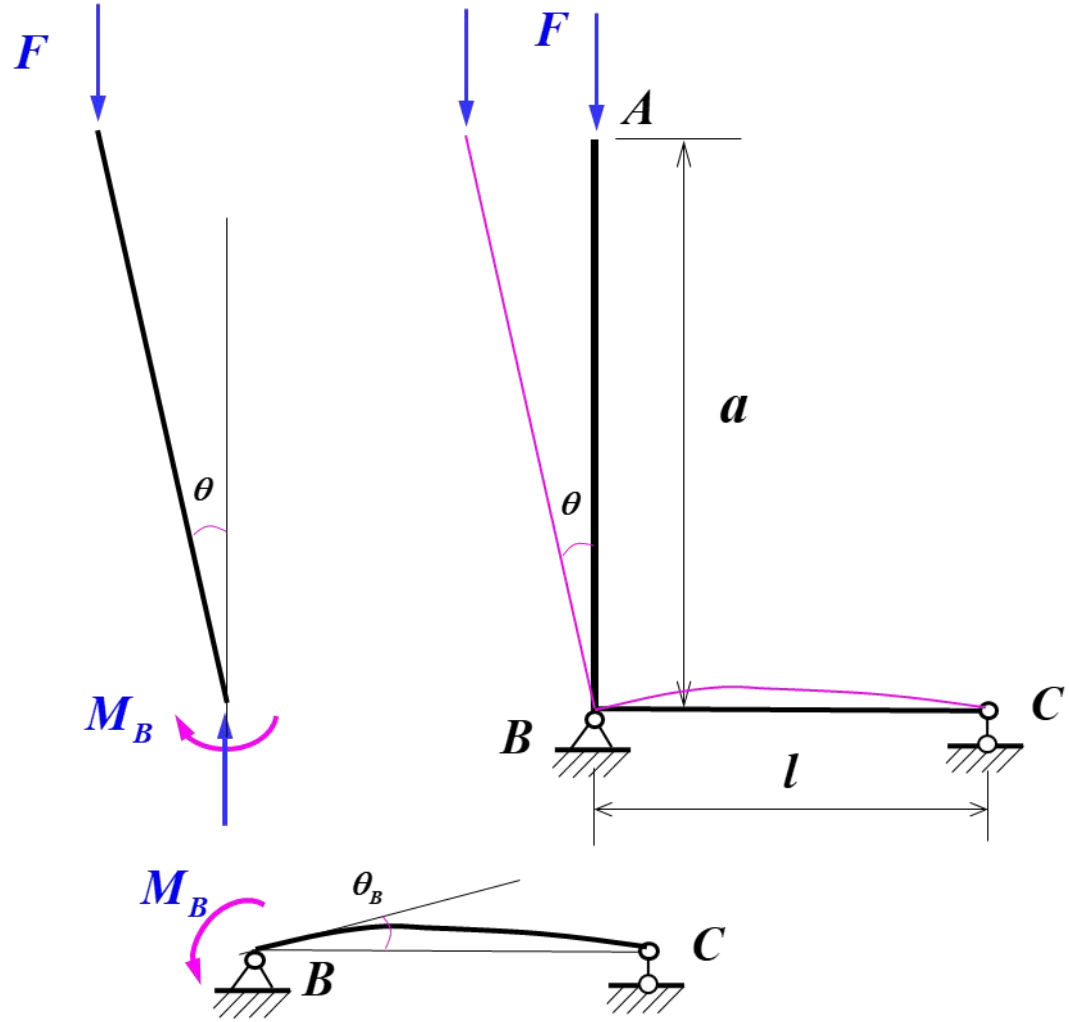
例11-15（习题11-4） 图示梁BC，EI为常数，B处竖直焊接刚性杆AB。求临界载荷。

解： 设刚杆偏转θ

$$Fa\theta = M_B$$

$$\theta = \theta_B = \frac{M_B l}{3EI}$$

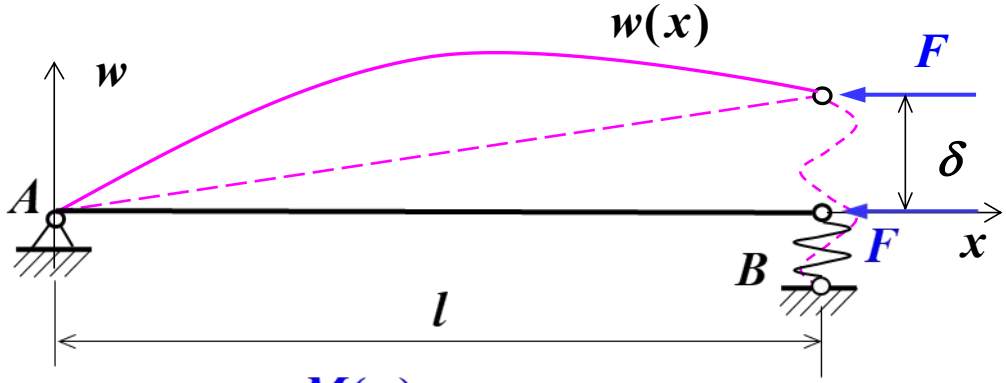
$$F_{cr} = \frac{3EI}{la}$$



3、结构的临界载荷

邻近状态的平衡

例11-16 图示梁AB，EI为常数，B处为弹簧支撑，弹簧刚度系数为c。求临界载荷。

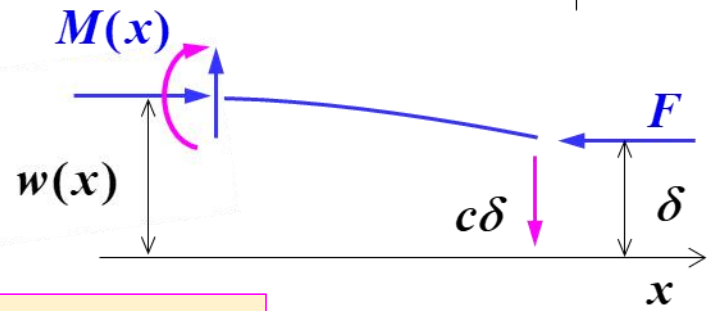


解法I

$$M(x)+F(w-\delta)+c\delta(l-x)=0$$

$$\frac{\mathrm{d}^2 w}{\mathrm{d}x^2}+k^2 w=\frac{c\delta}{EI}x,\quad k^2=\frac{F}{EI}$$

$$w(x)=A\sin kx+B\cos kx+\delta\frac{c}{k^2EI}x$$



$x=0,\ w=0;$

$x=l,\ w=\delta.$

$B=0$

$$A\sin kl+\delta\left(\frac{cl}{k^2EI}-1\right)=0$$

A,δ彼此独立，不同时为零

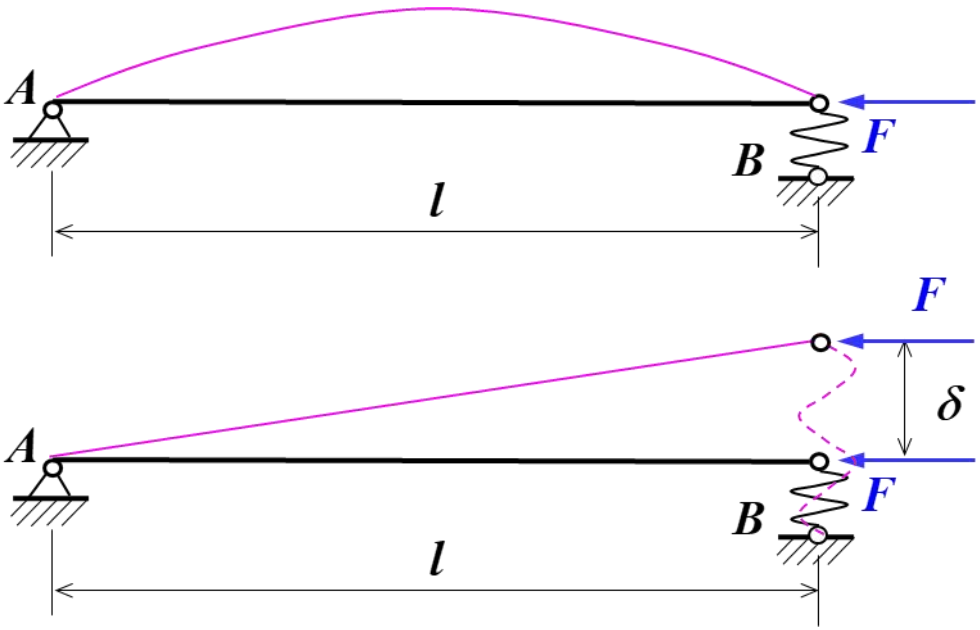
$$\sin kl=0,\quad F_{cr1}=\frac{\pi^2EI}{l^2}$$

$$\frac{cl}{k^2EI}-1=0,\quad F_{cr2}=cl$$

两种失稳模态；
刚度小者先失稳

3、结构的临界载荷

例11-16 图示梁AB，EI为常数，B处为弹簧支撑，弹簧刚度系数为 c 。求临界载荷。



解法II

考虑两种失稳形式

1）弹簧强，压杆失稳

$$F_{cr1}=\frac{\pi^2 EI}{l^2}$$

2）压杆强，弹簧失稳

$$F_{cr2}=cl$$

$$F_{cr1}=F_{cr2},\quad c=\frac{\pi^2 EI}{l^3}$$

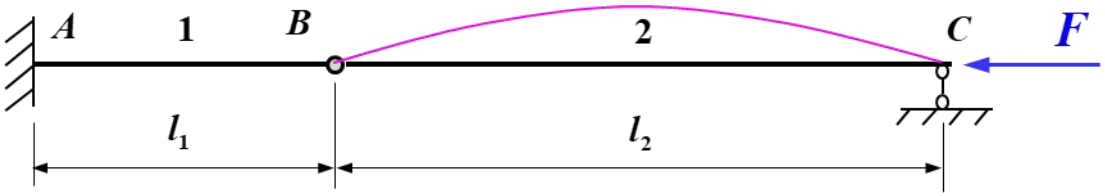
压杆弹簧同时失稳

3、结构的临界载荷

邻近状态的平衡

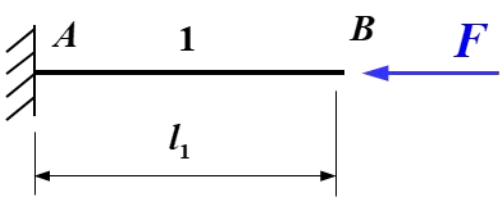
例11-17 图示细长杆1、2，EI相同且为常数，$l_2=2l_1=2l$。求临界载荷。

考虑两种失稳形式



1）假设1杆强，2杆失稳

$$F_{cr2}=\frac{\pi^2 EI}{l_2^2}=\frac{\pi^2 EI}{4l^2}$$

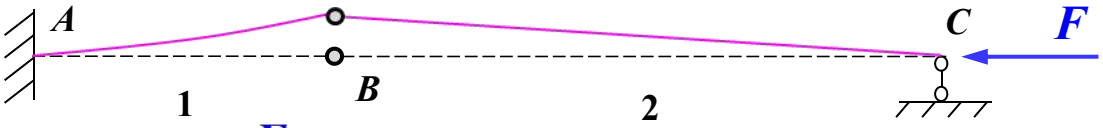


?

$$F_{cr1}=\frac{\pi^2 EI}{(2l_1)^2}=\frac{\pi^2 EI}{4l^2}$$

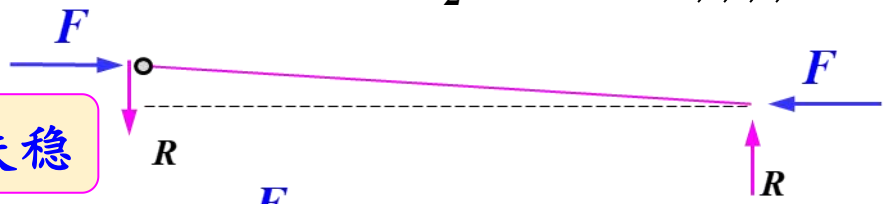
2）假设2杆强，1杆失稳

扰动导致的偏离

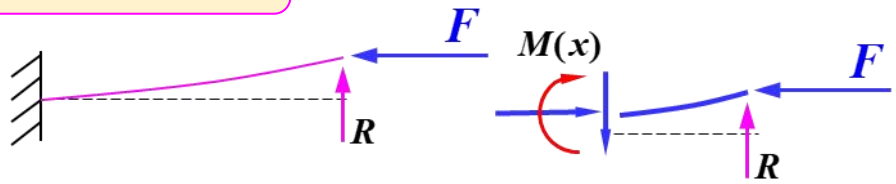


$$F_{cr1}<\frac{\pi^2 EI}{(2l_1)^2}=\frac{\pi^2 EI}{4l^2}$$

横向力促进失稳



$$F_{cr}=F_{cr1}=\frac{\pi^2 EI}{(2.37l_1)^2}$$



3、结构的临界载荷

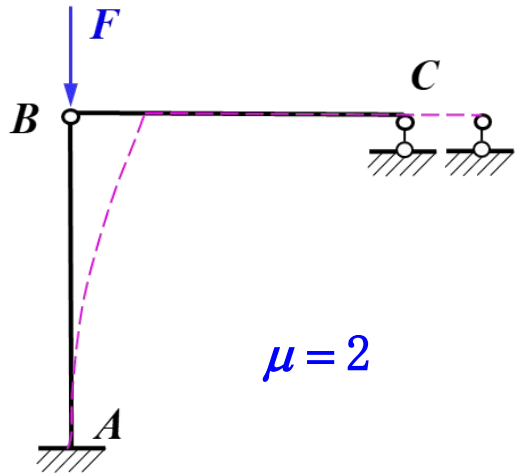
约束对杆件临界载荷的影响

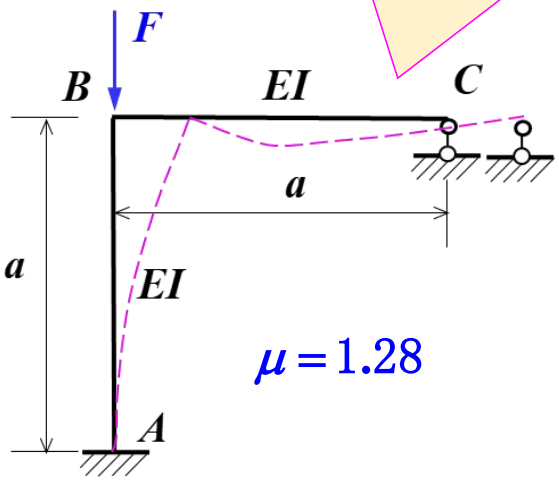
$$F_{cr}=\frac{\pi^2 EI}{(\mu a)^2}$$

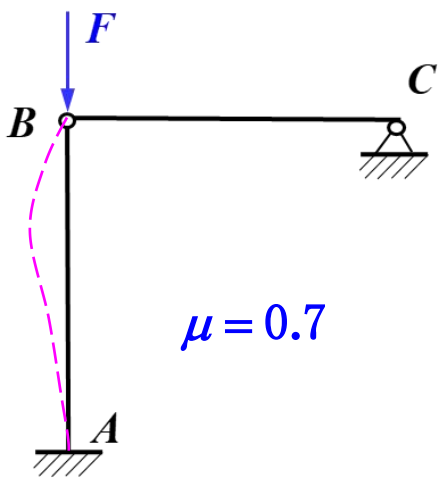
单辉祖，材料力学问题、例题与分析方法，高教社，2006；北航云盘

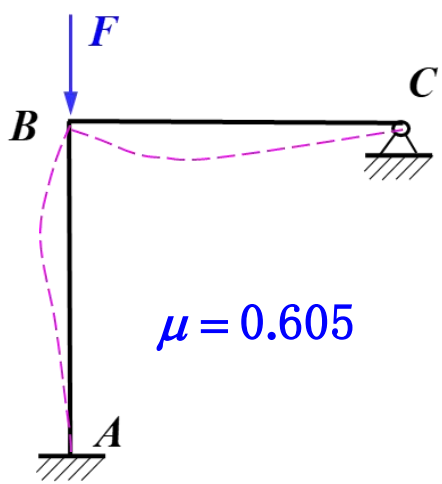
图示刚架，比较不同约束和连接形式的影响。

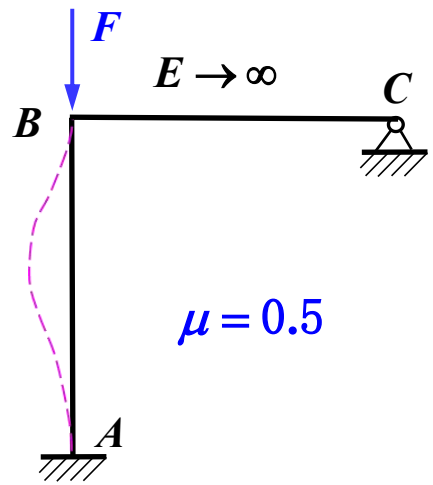
加强约束（平动、转动），限制弯曲变形，提高临界载荷



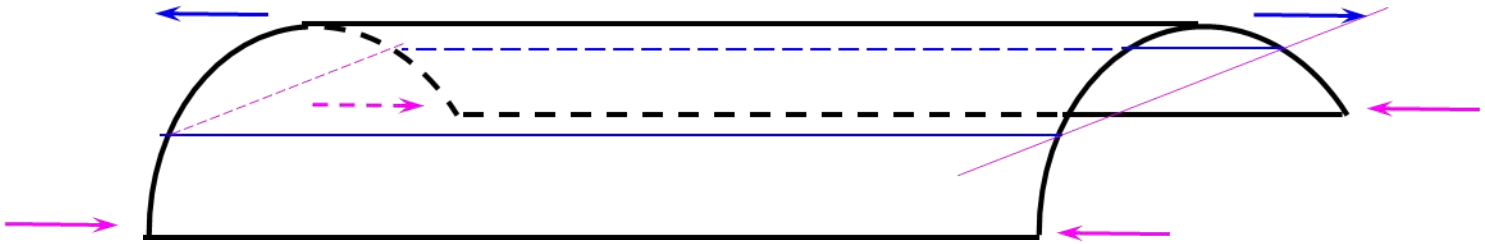


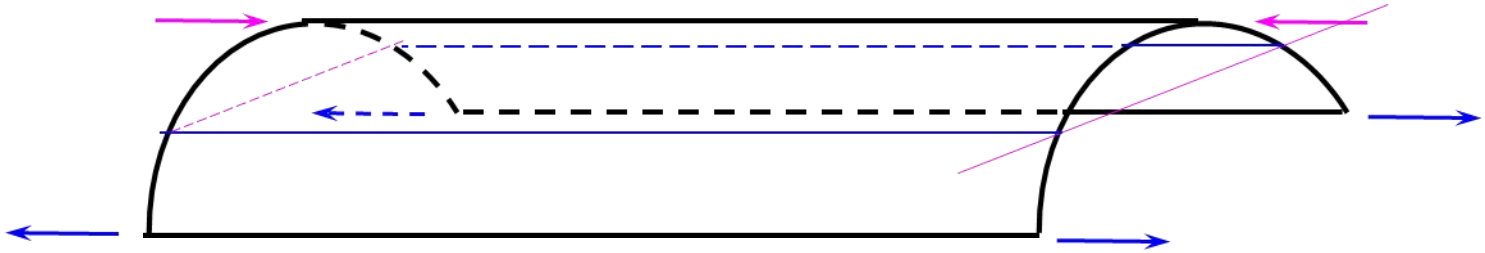






4、卷尺承载能力分析





塔吊臂受力分析

塔吊臂的截面形状

本讲：压杆稳定延伸与拓展

邻近状态的平衡

下讲：外力功与应变能